

EDUCATION

- **University of California, San Diego** La Jolla, CA, USA
Exchange Student, GPA: 4.0/4.0 Jan 2026 ~ June 2026
 - **Relevant Coursework:** Planning and Learning in Robotics (Graduate, A), Linear Control System Theory (A), Engineering Probability and Statistics (A+).
 - Tuition and stipend fully covered by the Korea-U.S. Student Exchange Program Scholarship.
- **Seoul National University** Seoul, South Korea
B.S. in Electrical and Computer Engineering, Summa cum laude, GPA: 3.89/4.0 Mar 2020 ~ Feb 2027
 - **Minor:** Mechanical Engineering.
 - **Relevant Coursework:** Robot Learning (Graduate, A+), Mechanical System Modeling and Control (A+), Introduction to Robotics (A+), Fundamentals of Control Engineering (A+), Mechatronics (A+).
 - Tuition and stipend fully covered by the Silla Scholarship Foundation.
 - Military leave of absence for 2 academic years (Feb 2022 ~ Nov 2023).
- **Daejeon Science High School for the Gifted** Daejeon, South Korea
Graduated with Honors; Specialized Curriculum in Mathematics and Physics Mar 2017 ~ Feb 2020

PUBLICATIONS

- [1] **Junsung Park***, Sachin Bhadang*, Carmelo Sferrazza, Sha Yi, and Xiaolong Wang. TACTX: Learning Shared Tactile Representations Across Diverse Sensors. **CoRL, 2026**.
- [2] **Junsung Park**, Hogun Kee, and Songhwai Oh. Modality-Augmented Fine-Tuning of Foundation Robot Policies for Cross Embodiment Manipulation on GR1 and G1. **RSS, 2026**, Workshop on Tactile Sensing for Robotic Foundation Models.
- [3] **Junsung Park**. BiCQL-ML: A Bi-Level Conservative Q-Learning Framework for Inverse Reinforcement Learning. **IEEE MIT URTC, 2025**.
- [4] Anya Zorin, Zilin Si, Myungsun Park, **Junsung Park**, Alexiy Buynitsky, Sachin Bhadang, Taejun Park, Sohee John Yoon, Yong-Lae Park, Oliver Kroemer, Zeynep Temel, Michael T. Tolley, Sha Yi, and Xiaolong Wang. TacO: Benchmarking Tactile Sensors for Object Manipulation. Preprint, 2026.

EXPERIENCE

- **Config** July 2026 ~ Present
Research Intern; Advisor: Prof. Kimin Lee
 - Researching methods to bridge the human-robot inverse dynamics modeling gap for robot action inference.
- **Wang Lab, UC San Diego** Jan 2026 ~ June 2026
Research Intern; Advisor: Prof. Xiaolong Wang
 - Developed a **sensor-agnostic tactile representation learning framework** for zero-shot policy transfer across heterogeneous tactile sensors.
 - Contributed to a **tactile sensing benchmark** for robotic manipulation, evaluating diverse sensors across complex contact-rich tasks.
- **Robot Learning Lab (RL LAB)** Jan 2025 ~ Dec 2025
Research Intern; Advisor: Prof. Songhwai Oh
 - Pioneered **modality-augmented VLA diffusion policies** for the **Unitree G1**, creating a **multi-modal dataset** that improved manipulation success from 48% to 94% via force integration; open-sourced the dataset on Hugging Face.
 - Engineered a **whole-body motion retargeting** pipeline converting the OMOMO dataset to robot kinematics, facilitating large-scale **imitation learning** of human interactions.
 - Implemented a **transformer-based semantic mapping** system that transforms egocentric RGB-D observations into allocentric **bird's-eye-view representations** for 3D spatial reasoning.
- **Robotics Laboratory** June 2025 ~ Dec 2025
Research Intern; Advisor: Prof. Frank Chongwoo Park
 - Developed and fine-tuned **vision-language-action models** by collecting demonstration data with the **Franka Emika Panda** and adapting the π_0 model for real-world policy transfer.
 - Developed a **physics-integrated action generation** framework that augments diffusion policies with a **Hamiltonian action encoder**, embedding **energy-conserving constraints** into the action manifold.

- **Soft Robotics and Bionics Lab**

Research Intern; Advisor: Prof. Yong Lae Park

June 2024 ~ Dec 2024

- Designed and prototyped a **wearable rehabilitation glove** integrating **Twisted String Actuators** and an **e-gain soft sensor** for assisted hand motion.
- Developed and integrated a motor-control architecture for real-time finger strain detection, enhanced by **IMU-based feedback control** to enable adaptive and **closed-loop motion regulation**.

CONFERENCE PRESENTATIONS

- **2025 IEEE MIT URTC**

Poster Presentation

Boston, MA, USA

Oct 2025

- “*BiCQL-ML: A Bi-Level Conservative Q-Learning Framework for Inverse Reinforcement Learning.*”

- **2025 ICCE-ASIA**

Poster Presentation

Busan, South Korea

Oct 2025

- “*BiCQL-ML: A Bi-Level Conservative Q-Learning Framework for Inverse Reinforcement Learning.*”

- **2024 SRRC Workshop**

Poster Presentation

Seoul, South Korea

Feb 2025

- “*Design and Implementation of a TSA-Driven Opposition Assist Glove with Integrated Soft Sensors.*”
- Presented a wearable rehabilitation glove integrating a rack-and-pinion actuator, e-gain soft sensor, and IMU-based feedback control for adaptive hand-motion assistance.

AWARDS AND HONORS

- **Gold Prize, SNU Creative Design Fair (out of 75 teams)**

Sep 2025

- Awarded a \$1,000 prize and \$4,000 in international conference support.
- Developed and validated a novel offline IRL reward-inference method through predictive-consistency evaluation on MuJoCo continuous-control benchmarks without online RL.

- **3rd Prize, Mechatronics Design Competition (out of 25 teams)**

June 2024

- Awarded a \$500 prize.
- Recognized for developing an innovative mechatronic system integrating mechanical design and control algorithms.

- **Korea-U.S. Student Exchange Program Scholarship**

Nov 2025

- Recognized for excellence in intelligent robotics research; awarded \$9,000 for outstanding academic and interdisciplinary achievement in AI-driven communication and control systems.

- **Semiconductor Track Scholarship**

Jan 2025 ~ Present

- Recognized for excellence in interdisciplinary studies; awarded \$4,300 to date, with \$10,000 to be awarded upon graduation.

- **Shilla Cultural Scholarship Foundation Scholarship**

Mar 2021 ~ Present

- Awarded full tuition coverage until graduation, recognizing academic excellence.

EXTRA CURRICULAR

- **SNU BUDDY (Cultural Exchange Organization for Exchange Students)**

Aug 2025 ~ Dec 2025

- Facilitated cultural exchange by introducing Korean culture and campus life to international exchange students.

- **Gongsang (SNU College of Engineering Student Reporters)**

Mar 2021 ~ Dec 2024

- Promoted STEM via web magazines on daily engineering, faculty insights, and engineering disciplines.
- Mentored high school students and supported outreach activities at the SNU Youth Engineering Frontier Camp.

PROGRAMMING SKILLS

- **Languages:** Korean (Native), English (Fluent)

- **Technologies:** Python, C++, MuJoCo, Isaac Sim, Isaac Gym, RoboSuite, MATLAB, Fusion 360, Simulink, Verilog